QM1 Statistics: week 2

Describing Distributions

Welcome!

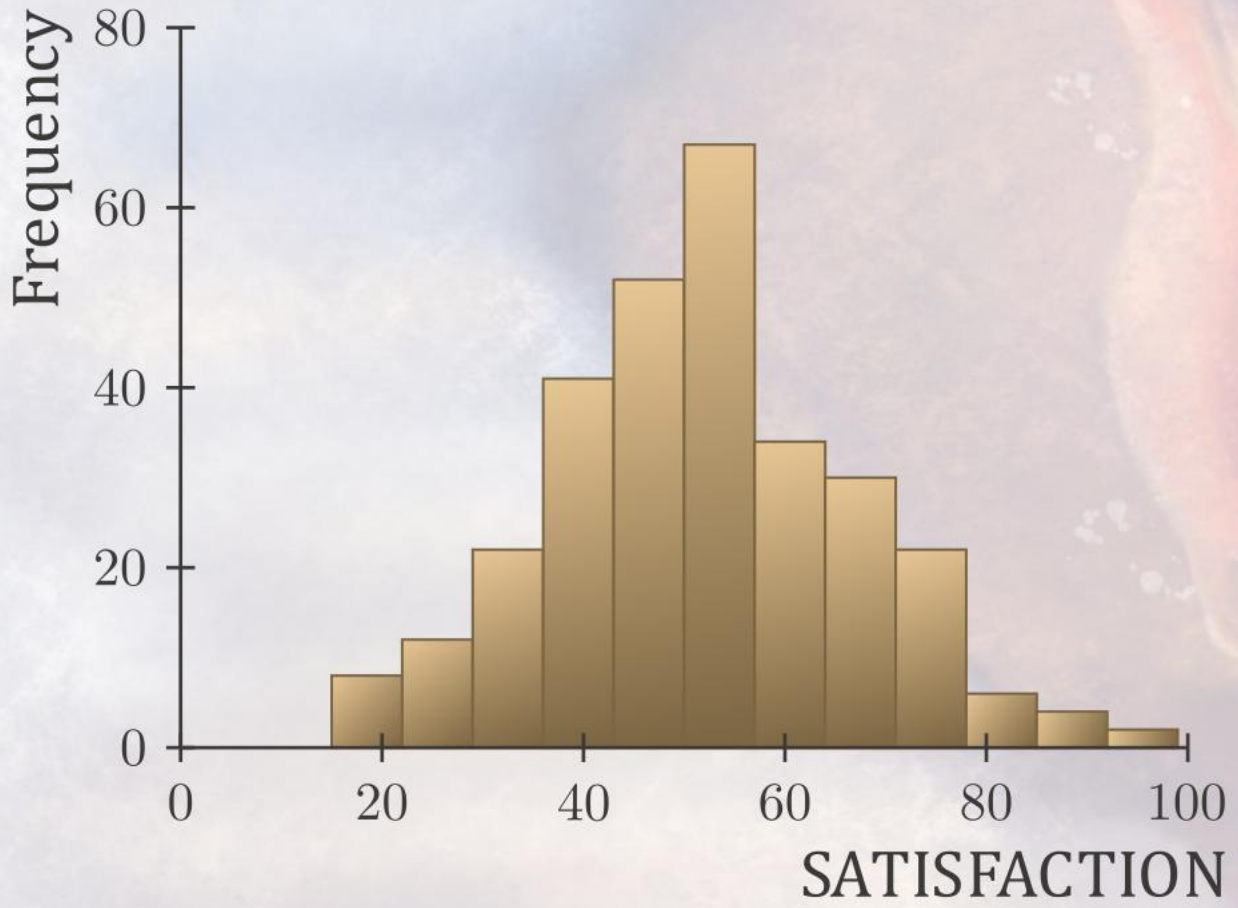
Perception is everything

Describing distributions

The way you look seems to become ever more important, also in societies deep beneath the sea

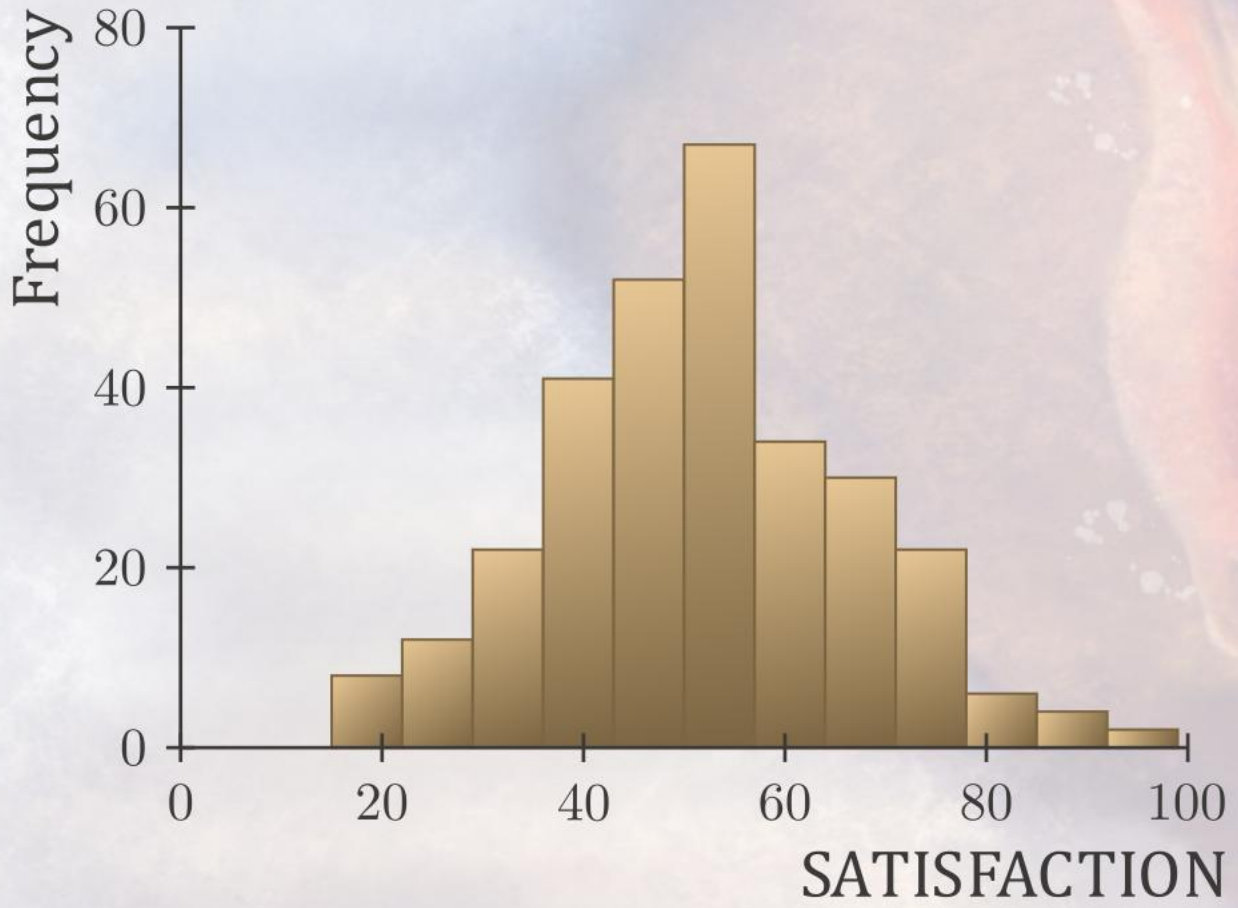
- Rumour has it that in the last decade,
  mermaids have opted for cosmetic surgery in increasing numbers
  (fin enhancements, et cetera)
- Research is needed. Therefore, sociologist Andersen (2023)
  asked 300 merpeople how they would rate their own body,
  on a scale from 0 ('monstrous') to 100 ('ouh-la-la')
  1. Observational unit: respondent (a merperson)
  2. Number of observations: n = 300
  3. Variable: body satisfaction, 0 – 100 (quantitative)

Say hi to the **histogram**



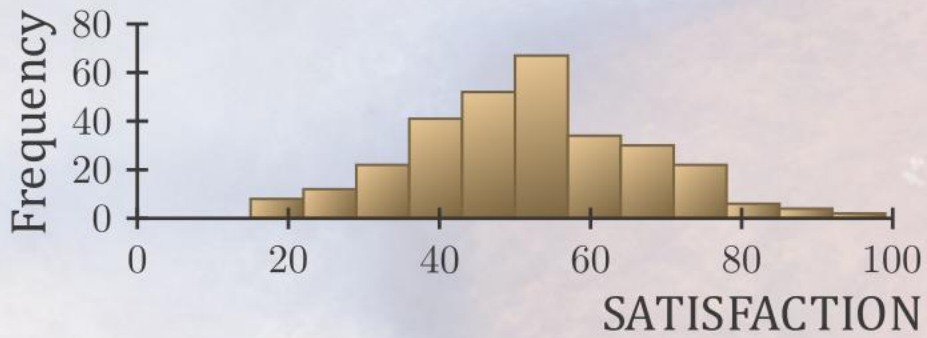
- The most popular graph for quantitative data
- We have 300 unique body scores…
- … and values that look alike are clustered into the same bar or 'class'
- We can thus see how the merpeoples' self-perceptions are generally divided, or 'distributed' along the scale
- You'll want to do 3 things:
  - I Identify the **shape**
  - II Locate the **centre**
  - III Capture the **spread**

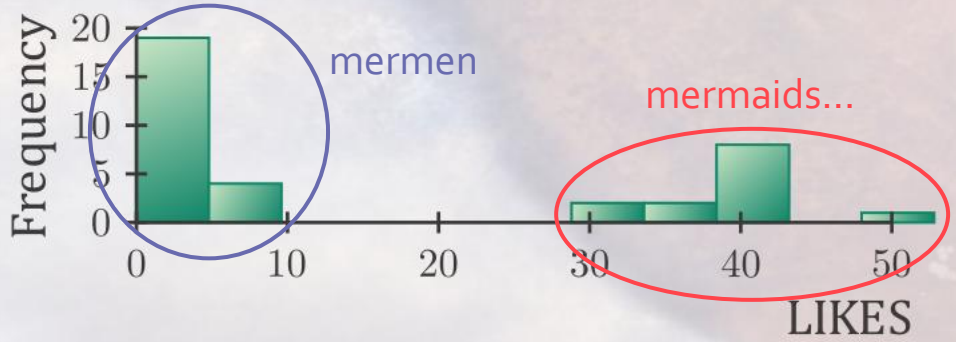
I The **shape** of a distribution

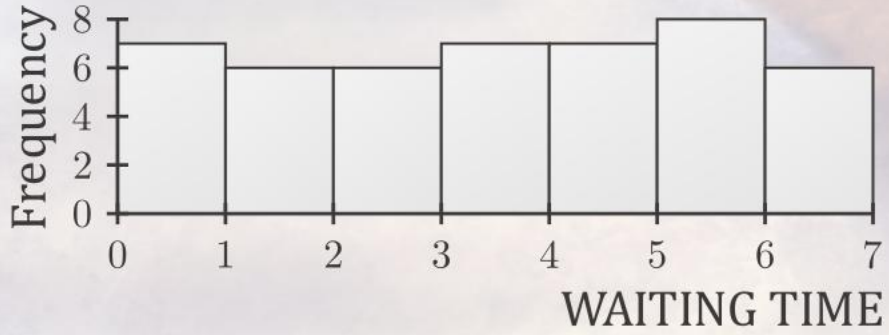


- Ask yourself 3 questions again:
  1. How many **peaks** (also known as **modes**) do I see?
  2. How **symmetric** is the graph?
  3. Does it contain **outliers**?

Shape, part 1: modes





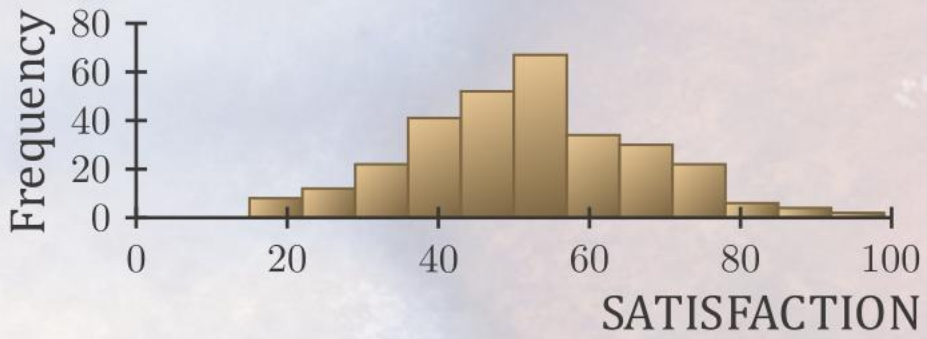


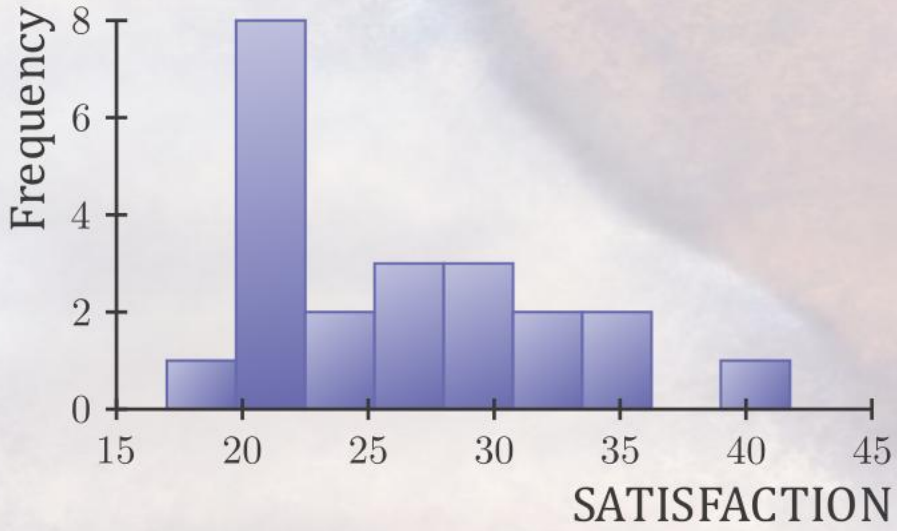
- **Unimodal** distribution
  - One peak
  - Nice and simple: people/products/… <u>tend towards something</u> ☺

- **Multimodal** distribution
  - Some respondents used a dating app (Blindr). Andersen asked them how many likes they had received last week
  - Could there be <u>multiple groups</u>? If so, why?

- **Uniform** distribution
  - The respondents who wanted cosmetic surgery could sign up at the clinic every Tuesday, so they didn't plan this in advance. Their waiting time varied between 0 and 6 days
  - No real peak

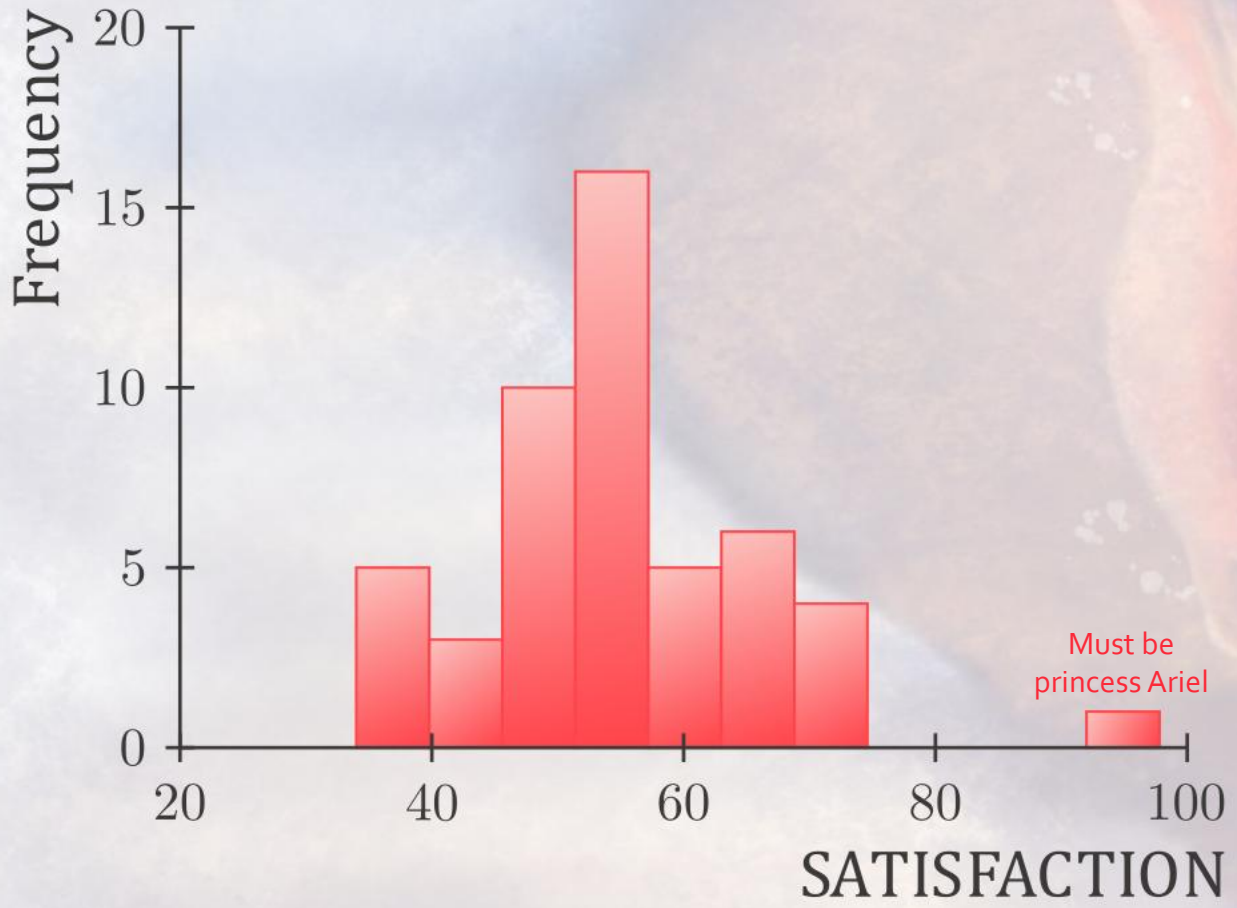
Shape, part 2: symmetry





- **Symmetric** distribution
  - A clear centre of gravity ☺
- **Skewed** distribution
  - When Andersen looked at the body satisfaction scores of only the male Blindr users… most of these were terrible, and a few were mediocre
  - This graph skews to the **right** (look at the 'tail')
  - Could be less straightforward to deal with (as we shall see)

Shape, part 3: outliers



- Among the girls under 20…
- … one score stood quite apart from the rest
- Outliers could be
  - Simple errors
    (fix them or throw them out!)
  - Genuinely unusual cases
    (can't just throw them out!)

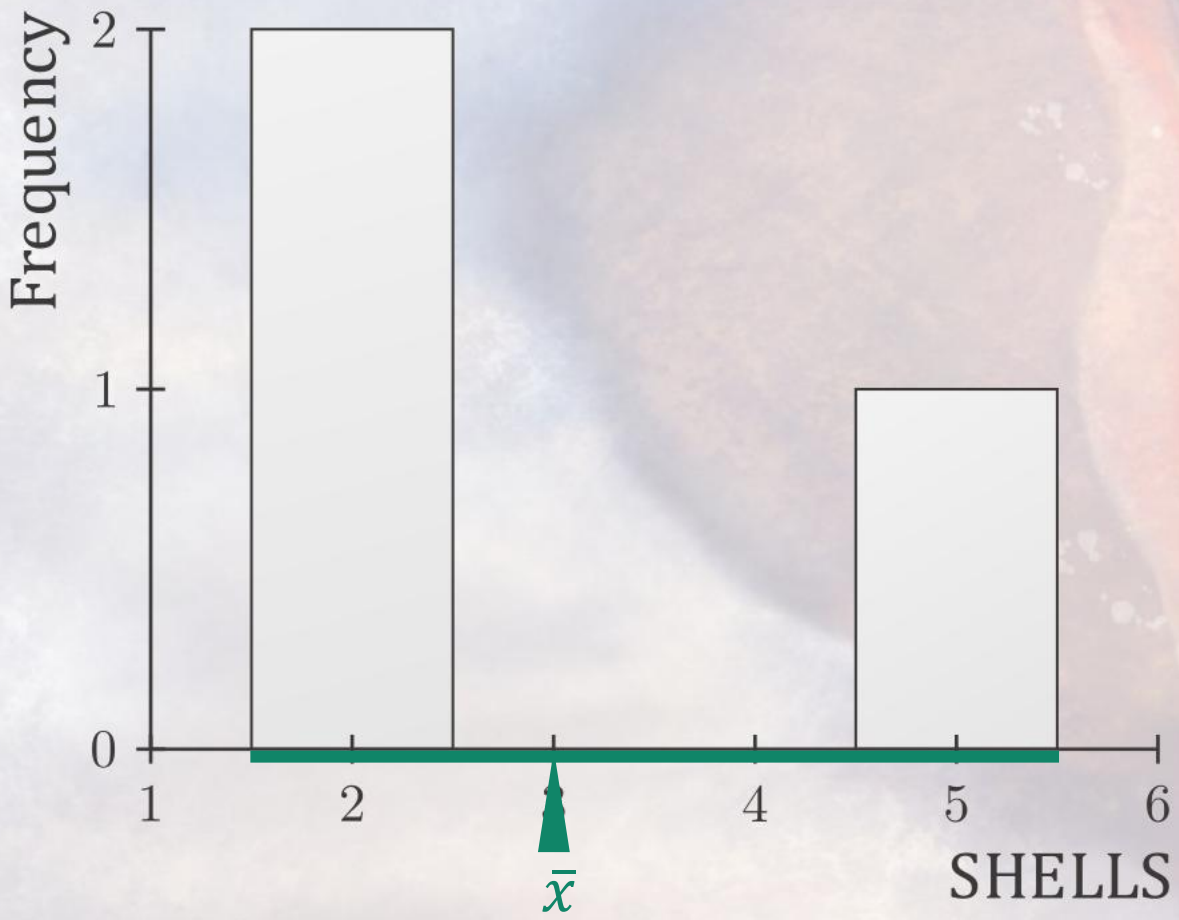
II The **centre** of a distribution

(British English FTW)

- What's the 'typical' case like? Choose the best statistic:
  1. The **mean**
  2. The **median**
  3. The **mode**

- To discuss each one, let's head on up to the beach, where a shell collecting contest is being held
  - Every contestant can hand in as many shells as they want
  - The prettiest seashell wins a prize

Centre, part 1: the **mean**



- Represents the average person/animal/thing/case… the **centre of gravity** of the distribution
- Small-scale example:
  - William hands in 2 seashells, Elizabeth does the same (2), and Hector offers up 5…
  - … so on average, these players submit

$$\bar{x} = \frac{2+2+5}{3} = 3 \text{ shells}$$

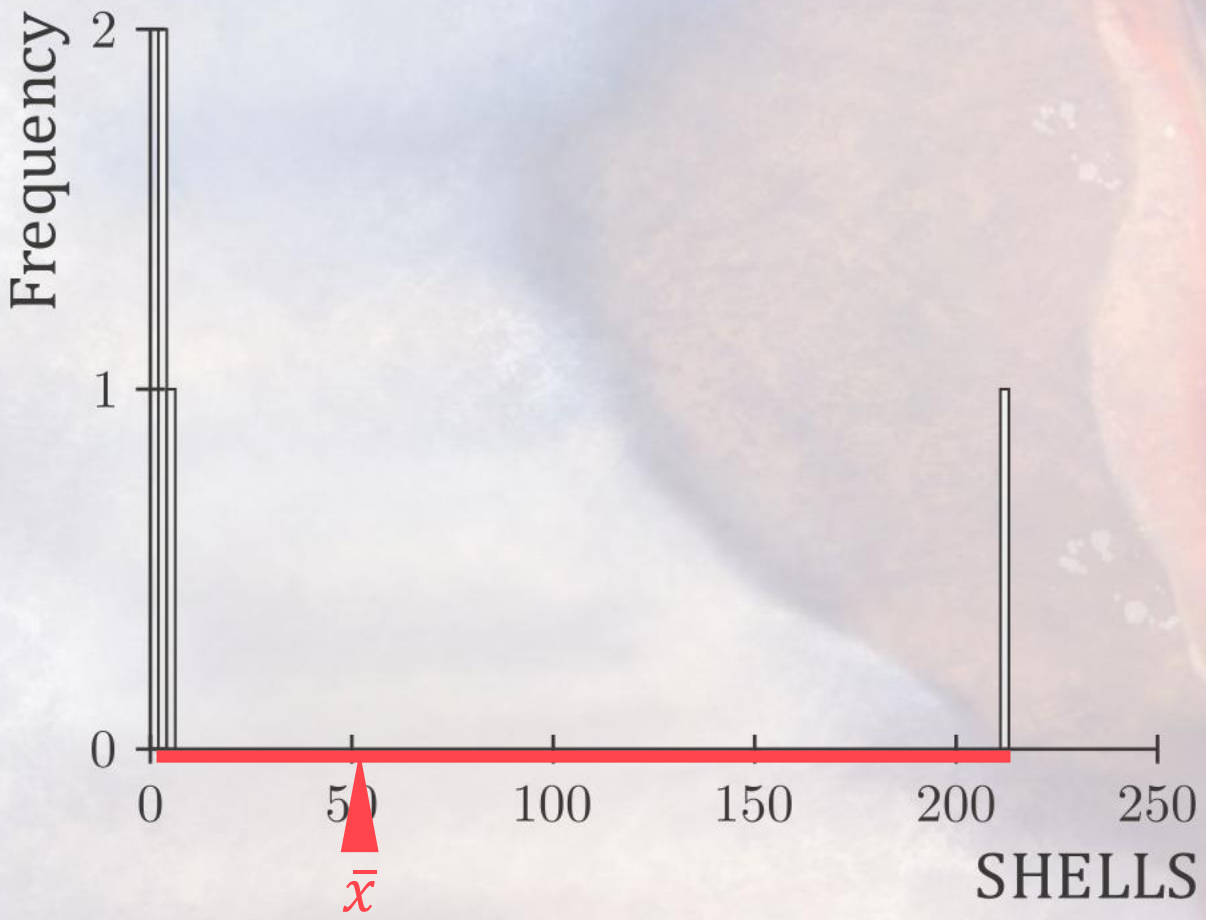
- General formula:

$$\bar{x} = \frac{\boxed{\sum x}}{\boxed{n}}$$

'sum all the *x*-values'

number of observations (a.k.a. 'sample size')

The mean is a perfect statistic with no flaws whatsoever



- … right?
- Unfortunately, the mean can be distorted by <u>skewness</u> and <u>outliers</u> ☹
  - After William (2 seashells), Elizabeth (2) and Hector (5), a fourth contestant walks in…
  - … and the nonchalant Jack empties a bag of 211 shells onto the jury table
  - The mean is now woefully unrepresentative:
    $$\bar{x} = \frac{2 + 2 + 5 + 211}{4} = 55$$
- The question is not:
  'Can I still *calculate* the mean?'
  (The answer is always 'yes')
- The question is:
  '*How much sense* does the mean make?'

Centre, part 2: the **median**

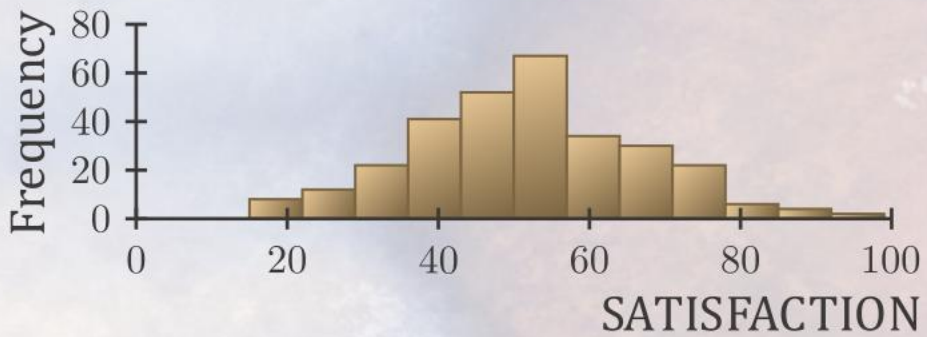
- Represents the <u>middle</u> person/animal/thing/case
- 50% lie under it, and 50% lie above
- Small-scale example:
  - Remember the horrible mean from the previous slide (55 shells)
  - The median, however, would be fine:

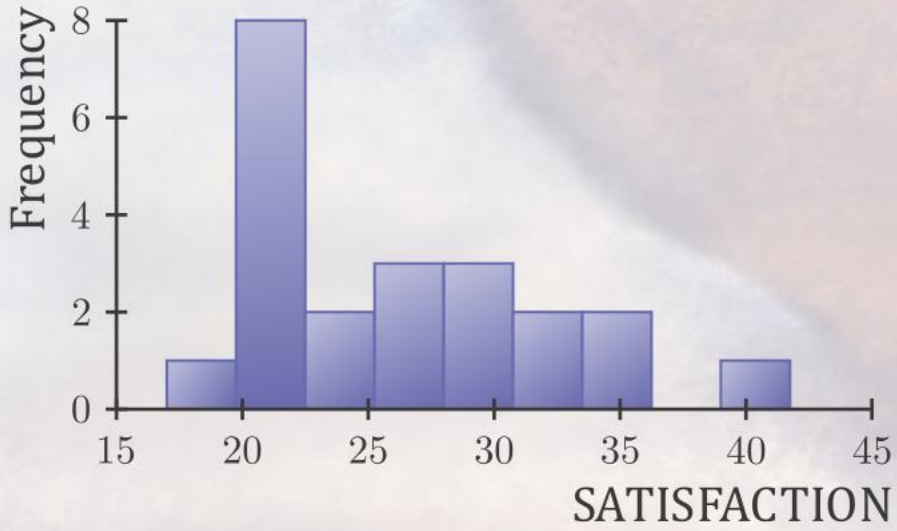
$$2 \quad 2 \quad 5 \quad 211$$

$$\boxed{M = 3.5}$$

  - Indeed: when *n* is even, take the <u>average</u> of the middle 2 cases as the median
  - Interpretation: the middle person submitted 3.5 seashells

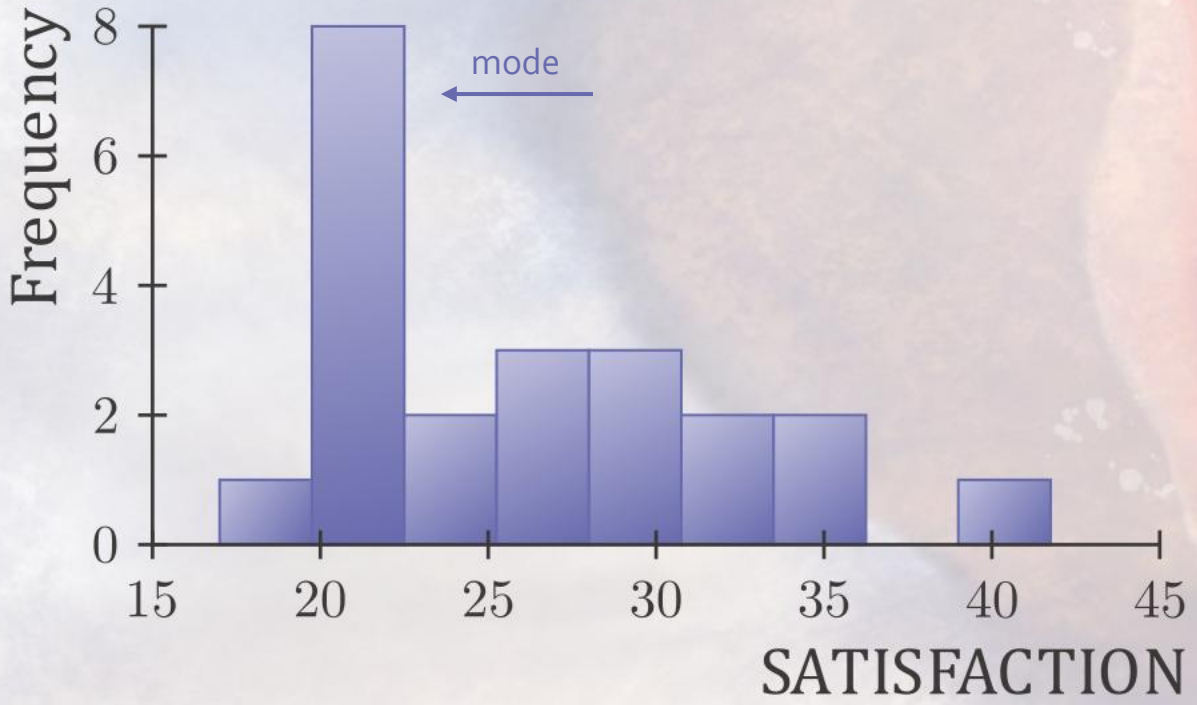
When to use the median?





- This statistic simplifies the 'information' in a variable quite a bit: it only looks at the one or two cases in the middle
  - Which makes it immune to outliers
  - But the mean accounts for the *numerical* information in *every* score… Isn't that usually nicer?
- Do you detect skewness or outliers? Keep using the mean if
  - The outliers have little impact on it (subjective!)
  - The skewness isn't all that extreme (subjective!)
- Switch to the median if
  - The outliers are clearly disruptive
  - The skewness is pretty awful

Centre, part 3: the **mode**

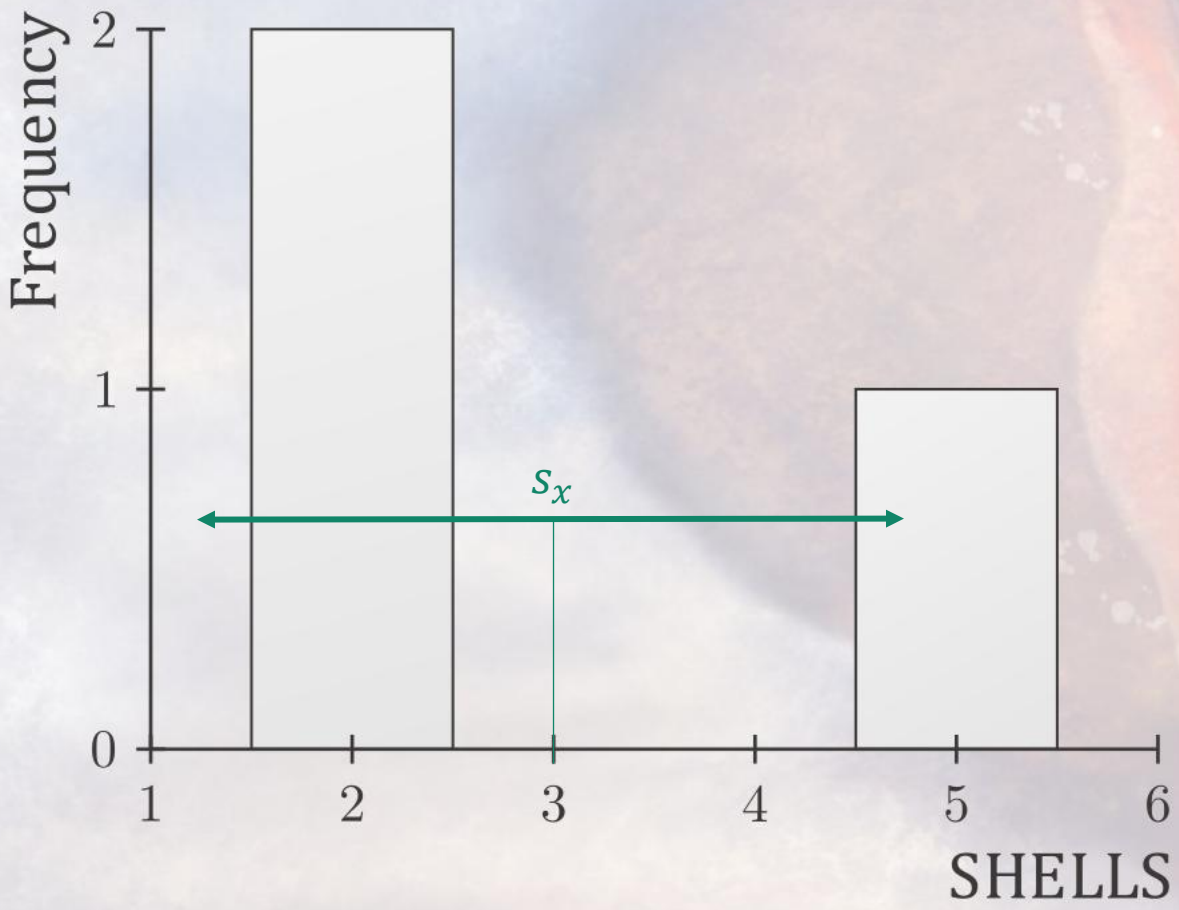


- Represents the person/animal/ thing/case that <u>occurs the most often</u>: the **peak** of the distribution
  - Can be a single value, but is often a certain *range* (since the histogram groups values together)
- A very basic statistic:
  - It ignores the numerical information in the values…
  - … and even their rank order
- Use only when your distribution is so skewed, even the median might not be representative

III The **spread** of a distribution

- How much do cases typically differ? Choose the best statistic:
  1. The **standard deviation**
  2. The **interquartile range**

Spread, part 1: the **standard deviation**

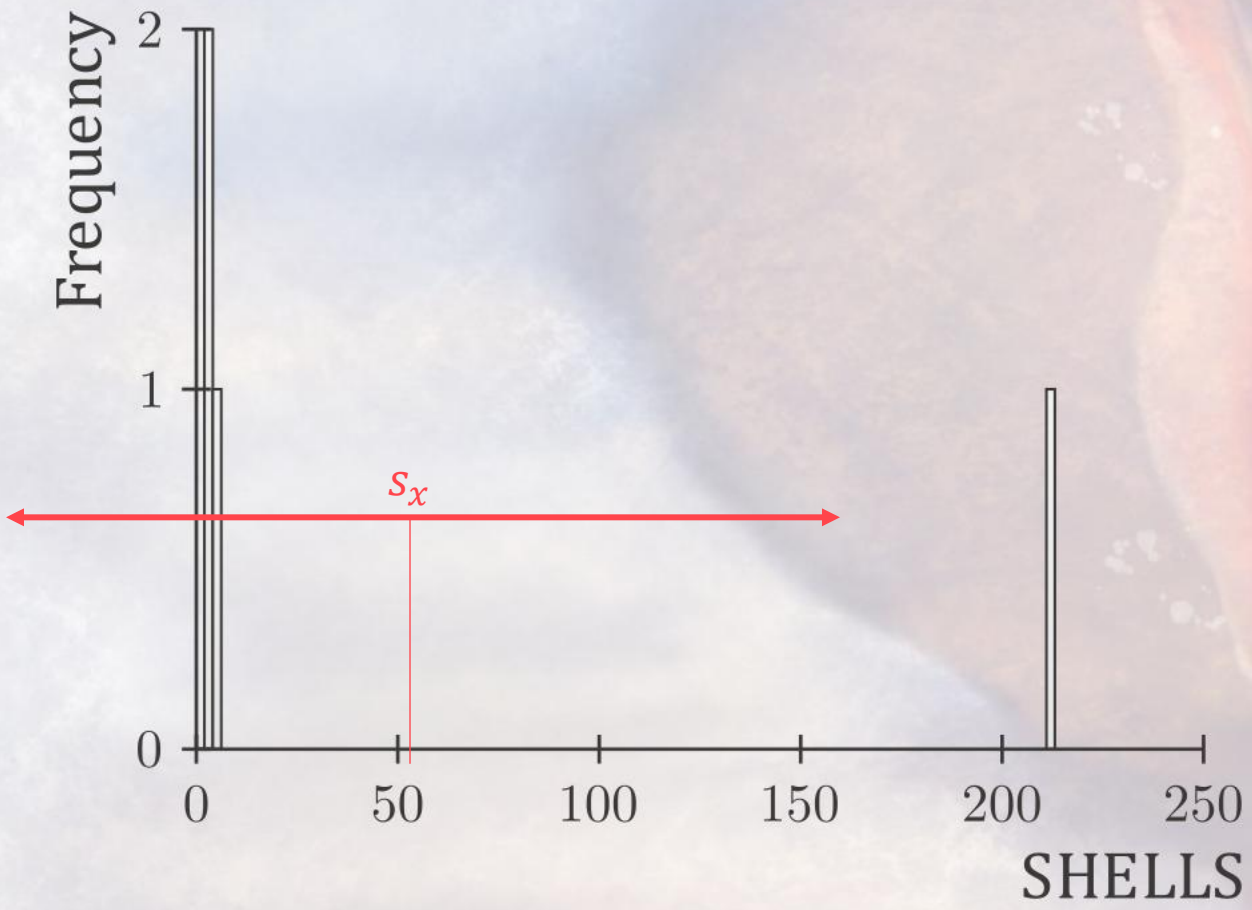


- The <u>average deviation from the mean</u>
- Returning to the contest:
  - William hands in 2 seashells, Elizabeth does the same (2), and Hector offers up 5…
  - … so the mean is 3, and the average player deviates from this mean by

$$s_x = 1.73 \text{ shells}$$

- The calculation is a bit icky, but probably useful to look at (see the book)

The standard deviation is a perfect statistic with no flaws whatsoever



- … Yeah, of course not. This thing can be distorted by <u>skewness</u> and <u>outliers</u> as well ☹
  - If we add Jack (211) to the group, the standard deviation turns into $s_x = 104.0$. Bugger! Way too big!
- Use the standard deviation if you have decided to use the mean ☺
- Otherwise, switch over to the interquartile range (see the next slide)

Spread, part 2: the **interquartile range**

- The jury gave n = 10 contestants the following grades
  (0 - 10 points for their prettiest seashell):
  - Anamaria: 9.0
  - Elizabeth: 8.0
  - Hector: 8.0
  - James: 8.0
  - Ian: 7.5
  - Josh: 7.5
  - William: 7.0
  - Jack: 7.0
  - Scarlett: 6.5
  - Davy Jones: 1.5

25%

$Q_3 = 8.0$

25%

$Q_2 = M = 7.5$

25%

$Q_1 = 7.0$

25%

IQR

- The IQR represents the <u>distance between the 50% cases in the middle</u>
- To find this, we need to make so-called **quartiles** first
- Next,

$$IQR = Q_3 - Q_1$$

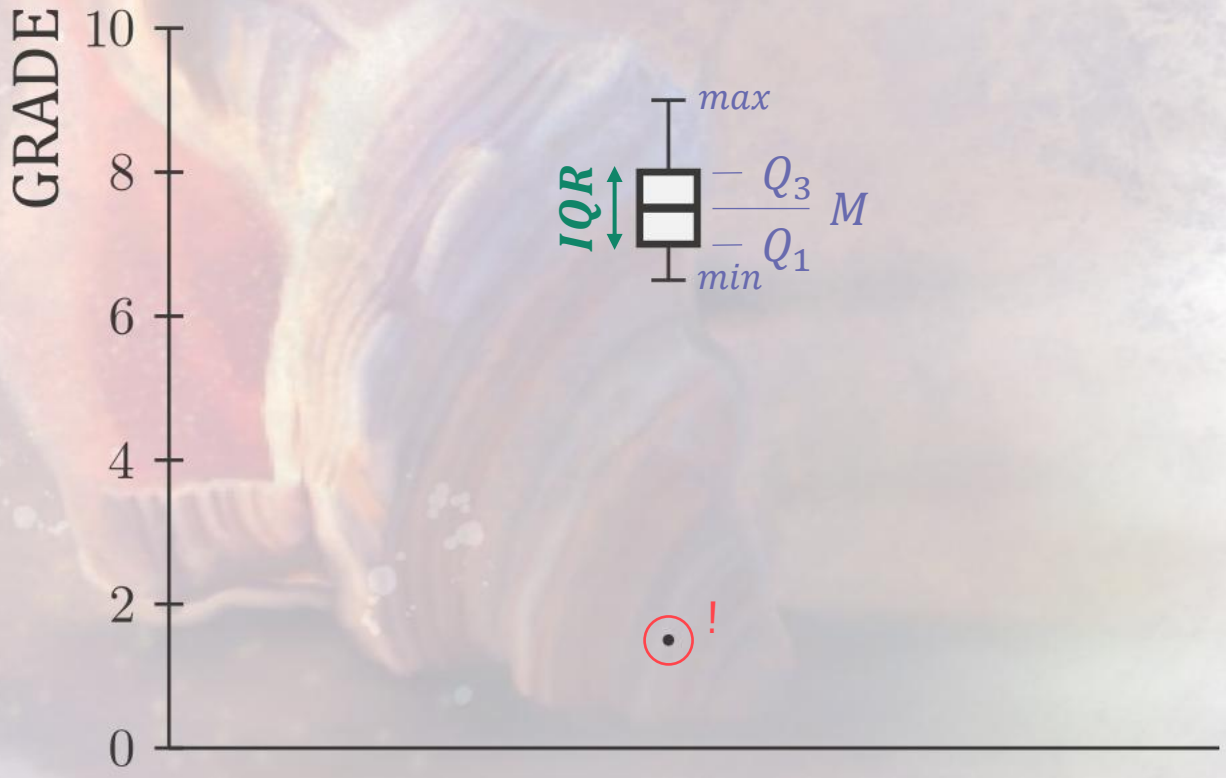
- So in the example on the left,

$$IQR = 8.0 - 7.0 = 1.0$$

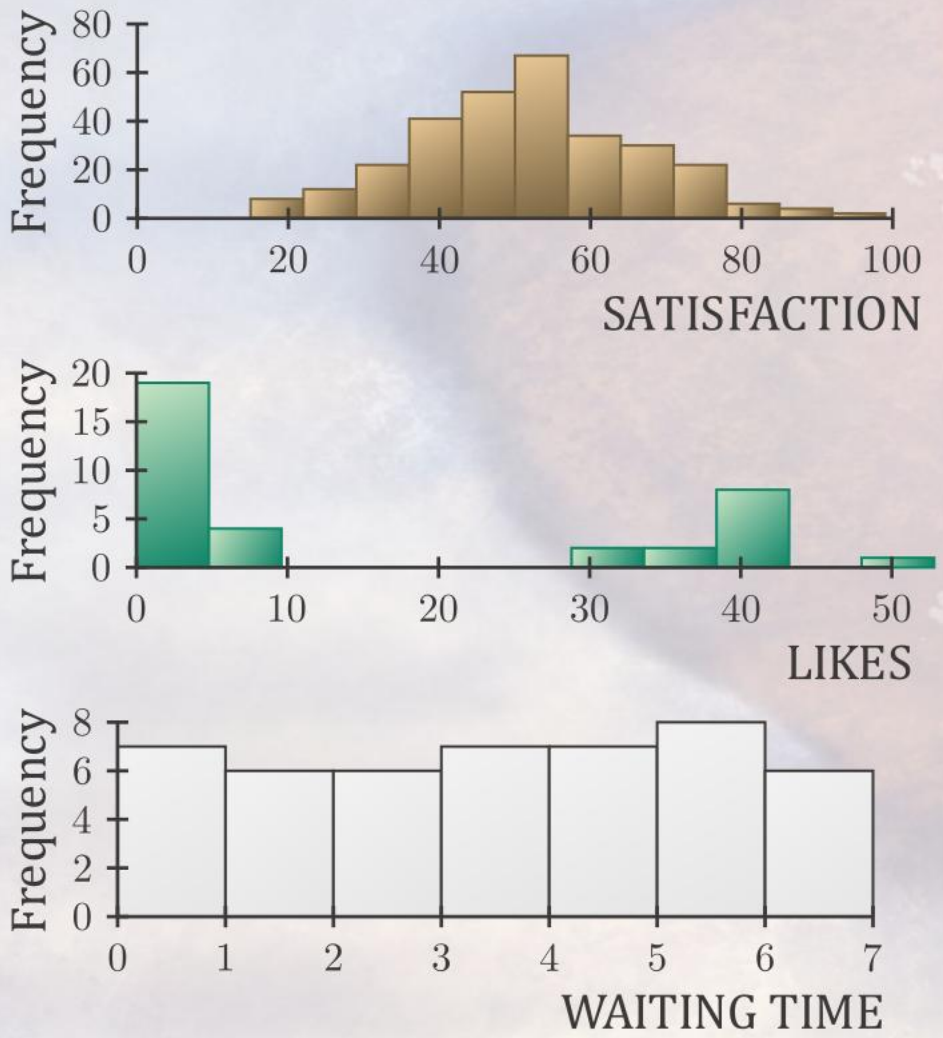
- Thus: half of the players (those in the middle) differ by 1 point or less
- If the IQR is large, the most 'regular' people are already far apart
- Note that the IQR **ignores** the outer 50%!
  - That's not ideal. On the other hand…
  - **Outliers make no impact on the IQR**

Visualising the IQR

- This **box plot** displays the quartiles
- Long box? Big IQR
- The so-called whiskers indicate the minimum and maximum
  - Outliers are excluded, though
  - On what basis would Davy Jones qualify as an outlier?
    - Check out the **1.5 IQRs rule** in the book!



Distributions, reprise

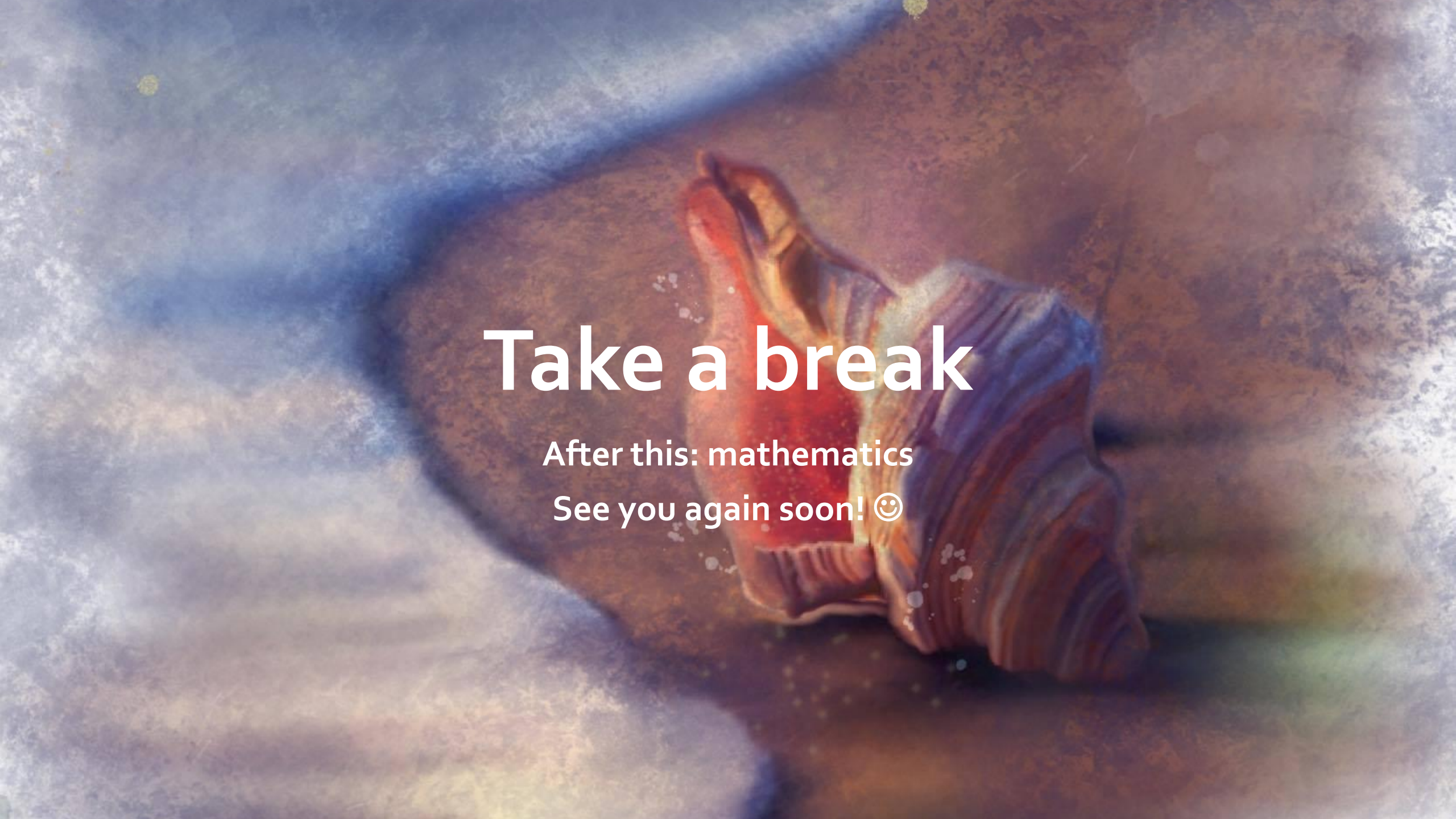


- Let's think back to our first question:
  *How many **modes** or **peaks** do I see?*
- Unimodal distribution:
  - Report **1** measure of centre and spread:
    - Mean + standard deviation, or
    - Median + IQR
      (in case of skewness and/or outliers), or
    - Mode + IQR
      (rare choice)
- Multimodal distribution:
  - Requires that you identify the subgroup variable ☺
  - Report a measure of centre and spread **per subgroup**
  - Stick to one choice (for example: median + IQR for all subgroups)
- Uniform distribution:
  - Report **1** measure of centre and spread (recommended: mean + standard deviation)
  - Mode makes no sense
    (and in case of perfect uniformity, mean = median)

Further topics for this week (self-study ☺)

- Sharpe, chapter 3
  - *Where's the Mode?* (info box on page 60)
  - How to calculate the standard deviation
  - Standardising variables
  - Five-number summary
  - The 1.5 IQRs rule for outliers (!)
  - Time series plots
  - Transforming skewed data

- Keep in mind: we expect you to read the whole book chapter. This slide highlights the most important aspects that were not covered in this lecture



Take a break

After this: mathematics

See you again soon! ☺